Rethinking the Arithmetic of Multi-Agent Medical Consultation

Anonymous ACL submission

Abstract

Multi-agent consultation has become the default design pattern in medical language-model systems, on the intuition that a panel of specialists diagnoses better than one physician. That intuition inherits an arithmetic premise from human medicine: pooling judgements helps because the judgements pooled are independent. Does it hold for language models? We measure the pairwise error correlation between panel members across 180 multi-agent configurations and 63,499 episodes, spanning three capability tiers, three medical benchmarks, five architectures and panel sizes $N \in \{1, 3, 5, 7, 9\}$. It is $\varphi = 0.73$, which leaves a nine-member panel with 1.31 **effective independent opinions**. Peer discussion raises the correlation to 0.986, so a six-agent consultation is informationally equivalent to 1.01 physicians at the moment it reaches consensus. Two consequences follow. Panel size predicts collaboration gain at cross-validated $R^2 = -0.006$, because tripling a panel adds a tenth of an opinion. And unanimity rises from 66% to 98% while accuracy given unanimity stays at chance, with confidence rising most on wrong answers, because agreement among copies reads as corroboration. The benefit that survives is confined to a window of task difficulty between 25% and 50% single-doctor accuracy, inside which all 20 significant improvements fall and outside which all 10 significant degradations do. Making medical panels agree is easy. Making them independent is the open problem.

1 Introduction

Ask six AI specialists a hard clinical question and let them confer. They will reach unanimity 98% of the time. They will be wrong half the times they agree. And by the moment they agree, they carry 1.01 independent opinions between them. Those three facts are one fact, and it is the subject of this paper.

Multi-agent "consultation" has become the default design pattern in medical language-model systems. MedAgents (Tang et al., 2024), MDAgents (Kim et al., 2024), MAC (Chen et al., 2025) and TeamMedAgents (Mishra et al., 2026) all instantiate the same clinical intuition. A panel of specialists should diagnose better than one generalist, the reasoning goes, because a tumour board diagnoses better than one physician. The pattern is being deployed faster than it is being measured.

The intuition is not merely an analogy; it is an inheritance. Pooling diagnostic judgements improves accuracy in human medicine for a specific reason, namely that physicians trained differently make different mistakes (Kurvers et al., 2016; Kämmer et al., 2017). Majority voting converts that independence into accuracy. Strip the independence away and the arithmetic stops working, however many voters remain. Nobody has checked whether language-model panels satisfy the premise they borrow.

They do not. Measuring the pairwise error correlation between panel members across 180 multi-agent configurations, we find $\varphi = 0.73$: agents prompted into different medical specialties fail on the same items. A nine-member panel therefore carries 1.31 effective independent opinions, and peer discussion drives the correlation to 0.986, at which point a six-agent consultation is informationally equivalent to 1.01 physicians. The panel does not converge on an answer. It collapses onto one.

Three observable consequences follow, and they are the same fact seen three ways. Panel size predicts nothing, because tripling the panel adds a tenth of an opinion. Consensus carries no information, because it is agreement among copies: unanimity reaches 98% while accuracy given unanimity stays at chance, and confidence rises most on the answers that are wrong. And the benefit that survives is confined to a narrow *window* of task difficulty rather than bounded by a ceiling, roughly 25–50% single-doctor accuracy, inside which every significant improvement we measure falls and outside which every significant degradation does.

Establishing this requires a grid large enough to separate architecture from panel size from task difficulty: 180 multi-agent configurations and 63,499 episodes across three capability tiers, three medical benchmarks, five canonical architectures, panel sizes $N \in \{1, 3, 5, 7, 9\}$, and a budget-matched single-model control at every cell. Medicine makes the measurement possible in two ways that general-domain agentic benchmarks cannot. Multiple-choice clinical reasoning uses no tools, so coordination can be measured without the confound of tool-call overhead. And expert-level items drive single-agent accuracy down to 15%, far enough into the difficult regime to expose the lower edge of the window, which is invisible at benchmark difficulties where single agents already succeed. Scaling principles for general-domain agent systems (Kim et al., 2026) supply the architecture taxonomy and the controlled-comparison design our grid is built on; the error-correlation and confidence-calibration measurements that carry our argument are new here.

2 Related work

Why pooled opinions help, and when they stop. Collective-intelligence results in medicine rest on error independence: pooling independent diagnostic judgements improves accuracy (Kurvers et al., 2016; Wolf et al., 2015), plateauing at roughly five raters (Kämmer et al., 2017), and groups of nine outperform individuals on the Human Dx platform (Barnett et al., 2019). This is the clinical instance of a general principle (Surowiecki, 2004) that multidisciplinary tumour boards institutionalise (Radcliffe et al., 2019). The framing question is not whether two heads beat one but when (Hautz and Kämmer, 2024). The same literature documents what happens when independence fails: conformity to a unanimous majority (Asch, 1956) and groupthink in deliberating groups (Janis, 1972) are the human analogues of the correlation collapse we measure. Correlated errors across models bound ensemble benefit in general (Kim et al., 2025), and analyses of multi-agent debate show consensus can converge on confident-but-wrong answers (Zhu et al., 2026). Diagnostic error is common enough in ordinary practice (Graber, 2013; Singh et al., 2014) that a consultation mechanism which manufactures unwarranted agreement is a safety concern rather than an efficiency one.

Medical language models and their evaluation. Medical question answering became a benchmark for general capability once models began matching licensing-exam thresholds (Kung et al., 2023; Nori et al., 2023), and Med-PaLM established expert-level performance as an explicit target (Singhal et al., 2023, 2025). Subsequent work moved from answering questions to conducting encounters, with AMIE demonstrating diagnostic dialogue and differential-diagnosis generation at physician level (Tu et al., 2025; McDuff et al., 2025). Prompting strategy alone carries much of this progress (Nori et al., 2024), which is precisely why architectural claims require budget-matched controls. A panel must be shown to beat the same model prompted harder, not merely to beat the same model prompted once.

Medical multi-agent systems. MedAgents (Tang et al., 2024) introduced role-played specialist collaboration; MDAgents (Kim et al., 2024) added adaptive routing between solo and group modes; MAC (Chen et al., 2025) and TeamMedAgents (Mishra et al., 2026) formalised teamwork components, the latter sweeping team sizes 2–5 under a Pareto-efficiency framing. Every published medical ablation stops at $N \leq 5$, none reports a dose–response curve, and none runs a budget-matched single-model control.

Skeptical evaluations. MedAgentBoard (Zhu et al., 2025) and a large npj Digital Medicine benchmarking study (Liu et al., 2026) independently report that multi-agent orchestration frequently fails to beat well-prompted single models on clinical tasks at substantially higher cost. ConfAgents (Zhao et al., 2025) quantifies the overhead at roughly $50\times$ the latency of a single generation for marginal gain. Our economics analysis extends this line quantitatively, and our independence measurement supplies the mechanism it lacks.

Multi-agent coordination in general domains. Debate-style protocols improve factuality on reasoning tasks (Du et al., 2024) and are argued to counteract degeneration of thought through enforced divergence (Liang et al., 2024); agent frameworks such as MetaGPT (Hong et al., 2024) and AutoGen (Wu et al., 2024) have made these patterns easy to deploy, and systematic studies compare collaboration mechanisms (Zhang et al., 2024). Kim et al. (2026) is the most complete treatment: a predictive scaling principle fitted across five architectures, three model families and six agentic benchmarks under matched budgets, from which the 45% capability ceiling and the error-amplification asymmetry are derived. We replicate its design in the medical, tool-free regime.

Scaling and test-time compute. Parameter and data scaling laws (Kaplan et al., 2020; Hoffmann et al., 2022) have analogues at inference time, where repeated sampling (Brown et al., 2024) and adaptive test-time compute (Snell et al., 2024) trade computation for accuracy. "More agents is all you need" (Li et al., 2024) reported monotone gains from ensemble size; Chen et al. (2024) showed the relationship is non-monotone, with an inverted-U in the number of LLM calls on hard instances. Li et al. (2026) confirmed the inverted-U in agent count, and Yang et al. (2026) attributed the fast-then-slow shape to opinion diversity, showing two diverse agents can match sixteen homogeneous ones. Self-consistency (Wang et al., 2023) over chain-of-thought (Wei et al., 2022) is the natural budget-matched control for any such claim, and is the control we run at every cell.

Confidence and calibration. Our safety analysis depends on whether a stated confidence is informative. Language models are partially aware of what they know (Kadavath et al., 2022), but elicited confidence is weakly calibrated and sensitive to how it is asked (Xiong et al., 2024), a failure mode with a long history in neural classifiers (Guo et al., 2017). Clinical evaluations report fluent but unreliable responses (Goodman et al., 2023), and demographic biases propagate through medical LLM outputs (Omiye et al., 2023; Zack et al., 2024). We add a coordination-specific finding: deliberation degrades the discriminative power of confidence rather than improving it.

3 Agent systems and tasks

3.1 System definition

We describe an agent system as a triple (A, C, Ω): a set of agents A, a communication graph C, and an aggregation rule Ω. Figure 1 and Table 1 give the five configurations we evaluate, each paired with the clinical role it instantiates. The taxonomy is the one used for general-domain agent systems (Kim et al., 2026); the clinical instantiations are ours. Only the coordination logic differs across architectures. The item rendering, the answer schema, the role prompts and the decoding parameters are shared.

Independent. N specialists receive the item and answer without seeing one another. Aggregation is majority vote, ties broken by mean self-reported confidence and then by option order. A confidence-weighted variant is logged throughout. This design gives the panel a single point at which errors can be caught before they reach the output. It is the architecture we find responsible for both the largest gain and the largest loss in the study.

Decentralized (MDT discussion). N specialists give first opinions, then exchange opinions on a complete graph for up to three rounds, each free to revise. Rounds stop early only on true unanimity, defined as every agent emitting the same *valid* option. A panel containing unparsed answers is not unanimous.

Hybrid (tiered referral). A generalist answers first with a stated confidence. Below a frozen threshold θ the case is referred to the N-specialist panel; if that panel has no majority the case escalates to full MDT discussion. θ is calibrated once on a 100-item development slice disjoint from every evaluation set and then frozen.

Panel composition. A cheap router (`gpt-4.1-nano`, temperature 0) ranks the nine most relevant specialties for each item from a fixed 17-specialty roster; a panel of size N is the first N of that ranking. Panels are therefore *nested*: the $N{=}9$ panel contains the $N{=}5$ panel, which contains the $N{=}3$ panel.

3.2 Nested panels as common random numbers

Every agent's first opinion is keyed on $(\text{item}, \text{specialty}_i, \text{seed}, i)$ and never on N or on the architecture. Consequently the first opinions of a panel of size N are a superset of those for any $N' < N$, and all four multi-agent architectures reuse the same first opinions. This is a deliberate common-random-numbers coupling. It reduces the variance of within-item contrasts between architectures and panel sizes, and it changes the estimand: we measure *accuracy*

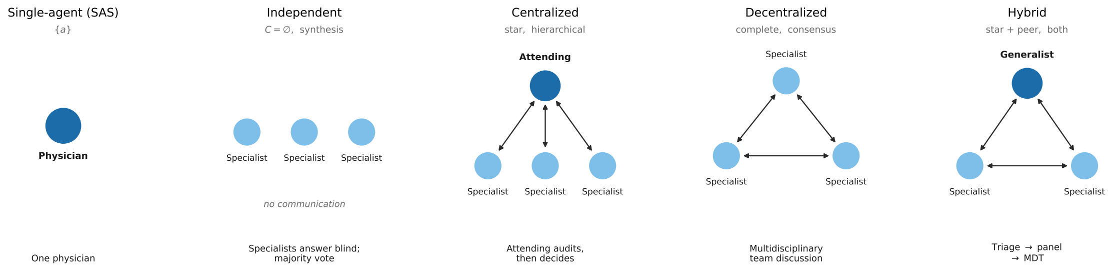

Figure 1: **The five architectures we evaluate**, with the clinical role each instantiates. A is the agent set, C the communication graph, Ω the aggregation rule.

Table 1: Architectures, their coordination structure, and the clinical role each instantiates. n is the panel size, k reasoning iterations, r orchestration rounds, d debate rounds.

Architecture	(A, C, Ω)	Cost	Clinical instantiation
Single-agent (SAS)	$\{a\}$	$O(k)$	One physician, zero-shot or CoT
Independent	$\{a_1..a_n\}$, $C = \emptyset$, synthesis	$O(nk)$	Specialists answer blind; majority vote
Centralized	$+a_{\text{orch}}$, star, hierarchical	$O(rnk)$	**Attending physician** audits, then decides
Decentralized	$\{a_1..a_n\}$, complete, consensus	$O(dnk)$	Multidisciplinary team discussion
Hybrid	$+a_{\text{orch}}$, star + peer, both	$O(rnk+pn)$	Tiered referral: triage $\to$ panel $\to$ MDT

as the same routed panel is extended, not the expected accuracy of independently redrawn panels at each size. Paired McNemar tests on item-level outcomes remain valid under this coupling; regression models must cluster at the item level rather than treat configurations as independent.

3.3 Clinical tasks and benchmarks

Three benchmarks, 250 items each, sampled with a fixed seed: MedXpertQA-Text (Zuo et al., 2025) (10 options; stratified by the 12 body systems $\times$ 3 task types), MedQA-USMLE (Jin et al., 2021) (4 options), and the contamination-screened hard split of MedAgentsBench (Tang et al., 2025) (894 items excluding the `medqa_5options` subset; the first 250 after stratified shuffling). The primary endpoint is exact match against the dataset's own gold option key. No LLM judge appears anywhere in the evaluation path.

Every item is additionally tagged easy / medium / hard by the pass rate of three mid-tier models at k=5 samples. On MedXpertQA the strata are heavily skewed hard (13 / 62 / 125 on the pilot sample), which is the intended regime.

4 Experiments and results

4.1 Setup

LLMs and capability scaling. Three OpenAI models form a capability ladder, measured rather than assumed (Table 11). The ladder is monotone on both benchmarks and spans single-doctor accuracy from 25% to 55% on MedXpertQA, straddling the previously reported 45% threshold from both sides. A task-grounded capability metric explains more variance than a generic intelligence index (Kim et al., 2026), so we take I to be each model's single-doctor chain-of-thought accuracy on MedQA.

Metrics and validation. We measure five coordination quantities. Their names are established in the agent-scaling literature (Kim et al., 2026), which reports headline values without always giving closed forms; the definitions below are the ones we compute.

- **Turns** T: reasoning–response exchanges; one model call is one turn.
- **Message density** c: peer-opinion exposures per turn. Independent has $C = \emptyset$ and therefore $c = 0$ by construction; Centralized exposes opinions only to the orchestrator; Decentralized and Hybrid expose every agent to every other each round.
- **Redundancy** R: mean pairwise Jaccard overlap of rationale tokens among first opinions. High R means the panel is restating one another rather than contributing independent information.
- **Overhead** $O\%$: token cost relative to the single-agent baseline on the same items,

$(\text{tok}_{\text{MAS}} - \text{tok}_{\text{SAS}})/\text{tok}_{\text{SAS}}$.

- **Coordination efficiency** E_c: accuracy divided by the token-cost ratio to the single-agent baseline.
- **Error absorption**: $(E_{\text{SAS}} - E_{\text{MAS}})/E_{\text{SAS}}$, where E is the error rate.
- **Error amplification** A_e: the extent to which individual-agent errors reach the final output without inter-agent verification. We measure that failure directly, in which the panel *held* the correct option among its first opinions and still emitted a wrong one, normalised by the marginal probability that a single agent is wrong. $A_e > 1$ means the system destroys information it already possessed.

Budget-matched controls. Every configuration is accompanied by three single-model controls. These are zero-shot, chain-of-thought, and **self-consistency at a matched budget**. The self-consistency pool is built from cached $n{=}8$ sampling calls (the API maximum), so any k is a prefix of the same samples. Nominal cost is charged as $\lceil k/8 \rceil$ prompts plus k completions, which is what an efficient implementation would actually pay. Panels are compared against the self-consistency cell whose mean per-item *dollar* cost is closest, not the one with a matching sample count. Matching on samples silently favours the panel, because a panel pays the prompt once per agent while self-consistency amortises it across a batch.

Cost accounting is *nominal* throughout. Cached calls are charged at list price, so the economics figures report what a configuration would cost to run, not what our cache made it cost.

Scaling law. We fit the functional form used for general-domain agent systems (Kim et al., 2026),

$$\begin{aligned} P = \beta_0 + \beta_1 I + \beta_2 I^2 + \beta_3 \log(1{+}T_{\text{tools}}) \\ + \beta_4 \log(1{+}n_a) + \beta_5 \log(1{+}O\%) \\ + \beta_6 c + \beta_7 R + \beta_8 E_c + \beta_9 \log(1{+}A_e) \\ + \beta_{10} P_{\text{SA}} + \textstyle\sum_j \beta_j (\text{interactions}), \end{aligned} \tag{1}$$

with interactions $I{\times}E_c$, $A_e{\times}P_{\text{SA}}$, $R{\times}n_a$, $c{\times}I$ and $P_{\text{SA}} \times \log(1{+}n_a)$. Our tasks use no tools, so $T_{\text{tools}} = 0$ identically and every term containing it vanishes. We fit the $T{=}0$ slice, which is the point of the study. Standard errors are clustered by benchmark, and cross-validated R^2 is reported over five folds.

Statistics. Paired McNemar tests with exact binomial p-values on identical item sets, Holm-corrected within each family of comparisons. Wilson 95% confidence intervals for all accuracies. Power was calibrated on a 200-item pilot. Median discordance between configurations is 8.5%, giving a minimum detectable difference of 5.7 percentage points at $n{=}200$ and 2.6 points at $n{=}1000$ with 80% power. Configurations that fail for infrastructure reasons (rate limits, timeouts) are recorded with an error status and excluded from accuracy. Scoring them as wrong answers would bias call-heavy architectures downward.

4.2 Main results

Across 180 multi-agent configurations and 63,499 episodes we find that collaboration in medicine is governed by task difficulty, not by panel size. Figure 6 gives the full dose–response surface; Table 10 the underlying numbers.

Collaboration pays only inside a window of task difficulty. Binning all 180 multi-agent configurations by the single-doctor baseline P_{SA} of their cell reveals a window rather than a ceiling (Fig. 7):

P_{SA}	configurations	mean gain	fraction positive
< 25%	40	−0.88 pp	45%
25–35%	40	**+2.29** pp	**92%**
35–50%	40	**+2.67** pp	**85%**
50–70%	20	−0.88 pp	50%
> 70%	40	+0.55 pp	68%

Configurations inside the 25–50% band gain +2.48 pp on average, while those outside it lose 0.31 pp (Welch $t = 8.42$, $p < 10^{-6}$, $n = 80$ vs 100). Splitting instead at the 45% figure of Kim et al. (2026) reproduces their direction but explains the data far less well (+1.36 pp below versus +0.07 pp above; $t = 3.46$, $p = 0.0007$). The previously reported ceiling is the window's *upper* edge. Its lower edge becomes visible only because expert-level medical items drive single-agent accuracy down to 15%. A window is not a weaker version of a ceiling. It identifies two distinct failure modes, and it describes the data better by a factor of 2.4 in t.

The window separates significant help from significant harm, with no overlap. Thirty of the 180 configurations differ significantly from the single doctor (McNemar $p < 0.05$), and they partition perfectly:

	sig. better	sig. worse
inside the window ($n = 80$)	**20**	**0**
outside the window ($n = 100$)	**0**	**10**

Every configuration that significantly beats the single doctor lies inside the window. Every configuration that significantly loses to it lies outside. The largest gain is the attending-physician (Centralized) panel at N=3: +7.60 pp at $P_{\mathrm{SA}} = 41.6\%$ (`gpt-5-mini`, $p = 2 \times 10^{-5}$), with +6.40 pp at $P_{\mathrm{SA}} = 32.0\%$ (`gpt-5-nano`, $p = 0.0037$). The largest loss is also a Centralized panel, at $P_{\mathrm{SA}} = 66.0\%$ with a weak model: -13.60 pp ($p = 10^{-5}$), where an orchestrator too weak to audit its own specialists overrides answers the panel had right. The window is not only a description of where gains concentrate. It is a two-sided boundary, and the harm on the far side is as significant as the benefit within.

Architecture matters more than panel size. Within the window, the ordering is stable: Centralized and Decentralized lead, Hybrid trails, and Independent voting is indistinguishable from the single doctor. On `gpt-5-nano` / MedXpertQA, Independent moves from 31.8% at N=1 to 32.6% at N=9, which is 0.8 pp for nine times the calls, while Decentralized reaches 38.0% at N=3 and stays there. Gains saturate at N=3 in every cell where they occur, and no architecture in any cell improves beyond N=5.

The best architecture changes with capability tier. The best architecture depends on the tier. At T2 the peak is Decentralized (38.0%, +6.0 pp); at T3 it is Centralized (49.2%, +7.6 pp). Family-dependent architectural preferences have been reported for general-domain agents (Kim et al., 2026), and the pattern here is the medical counterpart. The stronger model exploits an orchestrator's audit, whereas the weaker model benefits more from peer exchange, whose redundancy compensates for individually unreliable opinions.

4.3 Scaling principles

With accuracy as the response, the fit is degenerate. Fitting that form directly gives $R^2 = 0.997$ (5-fold cross-validated $R^2 = 0.996$), which is an artefact of our design rather than a result. The response is accuracy and one predictor is the single-doctor baseline P_{SA}; the same items and the same model underlie both terms and only the coordination structure varies, so P_{SA} alone nearly determines accuracy ($\hat{\beta}_{P_{\mathrm{SA}}} = 0.875$, $p < 10^{-4}$). We therefore refit with the collaboration *gain* as the response, which is what the scaling principle is actually about.

Coordination metrics explain the gain that accuracy alone cannot. With $\mathrm{gain} = P - P_{\mathrm{SA}}$ as the response, $R^2 = 0.518$ and 5-fold cross-validated $R^2 = 0.638$, with 5-fold cross-validated $R^2 = 0.415$. Coordination metrics carry that fit with no tool-count term at all. Coefficients (n = 225 configurations including the self-consistency control, benchmark-clustered SEs):

term	$\hat{\beta}$	p
$\log(1 + A_e)$	-0.150	< 0.001
R (redundancy)	$+0.082$	< 0.001
c (message density)	$+0.006$	< 0.001
$\log(1 + O\%)$	$+0.005$	0.024
$P_{\mathrm{SA}} \times \log(1+n_a)$	$+0.027$	0.048
$\log(1 + n_a)$	-0.003	0.701
P_{SA}	-0.113	0.051
P_{SA}^2	$+0.037$	0.390

Error amplification is the dominant penalty; redundancy and message density carry small positive effects; overhead is not penalised once T=0; and neither panel size nor P_{SA} has a main effect.

The tool–coordination trade-off is absent by construction, and coordination still predicts gain. $T_{\mathrm{tools}} = 0$ throughout, so the $\log(1 + T)$ main effect and the $O\% \times T$ and $E_c \times T$ interactions vanish identically. Coordination metrics remain highly significant without them, which establishes that the coordination effect is not a redescription of tool overhead. The coefficient on $\log(1 + O\%)$ turns *positive* here. With no tools to coordinate over, spending more tokens on deliberation is mildly beneficial rather than costly.

The decision boundary is not identifiable in the tool-free slice. The published threshold is derived as a coefficient ratio, $P^*_{\mathrm{SA}} = \hat{\beta}_4/\hat{\beta}_{17} \approx 0.063/0.408 = 0.154$ in standardised units, or ≈ 0.45 after denormalisation (Kim et al., 2026). In our fit $\hat{\beta}_{\log(1+n_a)} = +0.0005$ ($p = 0.95$) and $\hat{\beta}_{P_{\mathrm{SA}} \times \log(1+n_a)} = +0.0074$ ($p = 0.50$). The ratio, 0.064, rests on two null coefficients and carries no information. This is a statement about functional form, not a failed replication. A linear interaction between baseline and log panel size can express a monotone ceiling but cannot express a window. The binned analysis (§4.2) shows the relationship is non-monotone in P_{SA}, and a window is what the medical data require.

Capability enters non-linearly, but the quadratic term is not identifiable here. Our fit carries I and I^2 with large coefficients of opposite sign (-0.231 and $+0.228$), but bootstrap resampling ($n = 1000$) gives them standard errors of 0.94 and 0.54. With three capability tiers the quadratic is unidentifiable and we do not interpret it. Every other large coefficient is stable: $\log(1 + A_e)$ has bootstrap SE 0.024 and P_{SA} has 0.044.

The model is heteroscedastic, and we report robust standard errors. Residuals pass a normality test (Shapiro–Wilk $p = 0.207$) but fail homoscedasticity decisively (Breusch–Pagan $p = 0.0001$). All standard errors in this paper are therefore heteroscedasticity-robust (HC3), clustered by benchmark where more than two clusters exist. Residual standard error is $\hat{\sigma} = 0.017$, reflecting that collaboration gains in medicine are small in absolute terms. Regularised alternatives do not improve prediction: Lasso (10-fold CV for λ) retains 9 of 11 predictors at CV $R^2 = 0.382$ and Ridge reaches 0.144, both below the unregularised 0.415, so we retain the full specification.

Panel size alone has no predictive value whatsoever. Nested model comparison isolates where the explanatory power lives (5-fold CV R^2): the full model reaches 0.415; capability alone 0.196; coordination metrics alone 0.093; the single-doctor baseline alone 0.043; and **panel size alone** -0.006, worse than predicting the mean. Any account of medical consultation that treats "how many agents" as the design variable is fitting noise.

The scaling principle selects an architecture, given a case mix. The fitted model converts into a design rule. For a clinical deployment characterised by its single-doctor baseline P_{SA}, expected collaboration gain is positive only for $P_{\mathrm{SA}} \in [25\%, 50\%]$, is maximised by centralized coordination at $n_a = 3$, and is negative below 25%. Three archetypes make this concrete. *Board-style examination triage* ($P_{\mathrm{SA}} \approx 0.90$, MedQA-like): use a single model; no architecture we tested returns a significant gain and every panel costs 2–5× more. *Specialist referral review* ($P_{\mathrm{SA}} \approx 0.35$, MedXpertQA-like): use an attending-physician panel at $n_a = 3$; expected gain $+6$ to $+8$ pp at a $6.7\times$ cost premium. *Frontier diagnostic difficulty* ($P_{\mathrm{SA}} < 0.25$, MedAgentsBench-like at weak tiers): use a single model and spend the budget on a stronger one; panels lose 2.8 pp here because a panel of unreliable specialists reinforces its own errors.

Error amplification dominates the fit. $\log(1 + A_e)$ carries the largest coefficient of any coordination metric ($-0.150, p < 10^{-4}$). A_e measures how often a panel held the correct option among its first opinions and still emitted a wrong one. Configurations that discard information they already possess are exactly the configurations that fail to beat the single doctor. The same mechanism has been proposed from the opposite direction for general-domain agents (Kim et al., 2026); our measurement supplies the configuration-level evidence.

Redundancy provides benefit at scale. $R \times n_a$ is positive ($\hat{\beta} = +0.006$, $p < 0.001$), which matches the marginal-benefit-at-scale coefficient reported for general-domain agents (0.041, $p = 0.040$). In clinical panels, specialists restating one another is a weak form of verification rather than wasted capacity.

4.4 Coordination efficiency, error dynamics, and information transfer

Turn count grows sub-linearly with panel size. Fitting $T = a(n{+}0.5)^b$ gives $a = 0.85$, $b = 0.770$, $R^2 = 0.299$. The exponent is sub-linear, so turns grow more slowly than agents. A clinical panel answering a closed question terminates on unanimity rather than negotiating open-ended tool use. Turn counts by architecture are Decentralized 7.5, Centralized 6.5, Independent 5.0, Hybrid 1.9, against a single-agent 1.4, a range of 1.4 to $5.3\times$. The hard resource ceiling they infer from super-linear growth does not arise in tool-free consultation.

Message density does not predict accuracy. $S = 0.462 - 0.045 \ln(c)$, $R^2 = 0.056$: message density explains almost none of the variance in accuracy. Our Centralized panels run at $c = 0.99$ and Decentralized at $c = 0.79$, well past the $c^* = 0.39$ plateau reported elsewhere, with no accuracy penalty. Once a panel is communicating at all, communicating more neither helps nor hurts.

Error absorption is real but small. Absorption $(E_{\mathrm{SAS}} - E_{\mathrm{MAS}})/E_{\mathrm{SAS}}$ is positive for every architecture: Decentralized $+4.2\%$, Hybrid $+2.7\%$, Independent $+1.9\%$, Centralized $+1.6\%$. The published figure is 22.7% (95% CI $[20.1, 25.3]$) for verification-bearing architectures. Medical panels

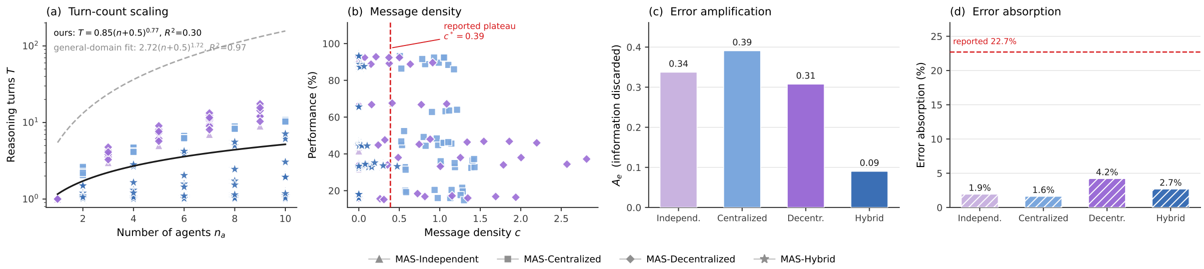

Figure 2: **Coordination dynamics.** (a) Turn count against agent count, with our fit and the published general-domain power law for comparison. (b) Performance against message density, with a reference plateau marked at $c^* = 0.39$. (c) Error amplification A_e and error absorption by architecture.

absorb roughly a fifth as much error as general-domain agent teams do, and the ordering differs: peer discussion absorbs most, not orchestration.

Coordination failure is structurally impossible without a channel. We classify first-round errors into four categories, following the multi-agent failure taxonomy of Cemri et al. (2025):

	Indep.	Centr.	Decentr.	Hybrid
Logical contradiction	6.3%	6.3%	6.3%	3.4%
Numerical drift	4.5%	4.7%	4.8%	3.1%
Context omission	13.0%	13.3%	13.4%	6.8%
Coordination failure	**0.0%**	3.0%	1.2%	0.0%

The structural prediction replicates exactly: coordination failure is **0.0%** under Independent coordination, for a structural reason: with $C = \emptyset$ there is no channel on which coordination can fail. Hybrid is also at 0.0% because its triage gate declines to convene a panel on most items. The magnitudes do not replicate: peer verification is reported to cut logical contradiction from 16.8% to 11.5%, whereas our first-round contradiction rate is flat at 6.3% across communicating and non-communicating architectures. Numerical drift is an order of magnitude rarer than in tool-using tasks (3–5% against 19–26%), as expected when no arithmetic is cascaded.

Information gain is positive but does not buy accuracy. Discussion reduces the entropy of the panel's answer distribution by $\Delta\mathcal{I} = +0.90$ bits (Decentralized), $+1.46$ bits (Hybrid) and $+0.24$ bits (Centralized) between the first and final round. Panels converge, and they converge most where the architecture communicates most. Convergence occurs whether or not the consensus is correct: $\Delta\mathcal{I}$ does not predict collaboration gain ($r = 0.055$, $p = 0.59$). Medical consultation is information-limited rather than information-transferring.

Architecture rankings do not generalise across benchmarks. Kendall's τ between the architecture rankings of any two benchmarks averages $+0.067$ (pairwise -0.200, $+0.400$, 0.000; none significant). Leave-one-benchmark-out cross-validation gives $R^2 = -0.015$: a model fitted on two medical benchmarks predicts the third no better than the mean. Coordination principles that transcend task structure in general-domain agentic work do not transcend benchmark in medicine. This is the practical counterpart of the window: architecture must be selected per difficulty regime, and difficulty regime does not coincide with domain.

The architecture that wins on accuracy loses on tokens by $4.3\times$. Successes per 1,000 tokens: Hybrid 0.80, self-consistency 0.47, Independent 0.39, Decentralized 0.33, Centralized 0.19. Centralized is $4.3\times$ less token-efficient than the cheapest configuration. The ordering closely matches theirs (SAS 67.7; Decentralized 23.9, $2.8\times$; Centralized 21.5, $3.1\times$; Hybrid 13.6, $5.0\times$), with one inversion. Hybrid is the *most* efficient architecture in medicine, because its triage gate declines to escalate most cases and therefore rarely pays for a panel at all.

Panels beat budget-matched self-consistency only inside the window. On MedXpertQA at `gpt-5-nano`, Decentralized at $N{=}3$ reaches 38.0% while self-consistency at the closest matched dollar cost reaches 36.0%. Outside the window the control wins or ties: on MedQA at the same tier, self-consistency at $k{=}15$ (88.4%) is within noise of the best panel (89.6%) at lower cost, and on MedAgentsBench at T1 self-consistency (20.0%) beats every architecture. Coordination buys something beyond repetition, and only where the window says it should.

Table 2: Effective independent opinions. φ is the mean pairwise error correlation between panel positions; $N_{\text{eff}} = N/(1 + (N-1)\varphi)$. Hybrid is lower because its triage gate declines to convene a panel on most items, so its measured pairs are drawn from the harder subset it does escalate.

Architecture	N	φ	N_{eff}	information used
Independent	3	0.743	1.21	40%
	5	0.731	1.27	25%
	9	0.734	1.31	15%
Centralized	9	0.735	1.31	15%
Decentralized	9	0.735	1.31	15%
Hybrid	9	0.493	1.82	20%

Table 3: Clinical safety. Unanimity is measured on the last round with two or more agents; $P(\text{wr.}\,|\,\text{un.})$ is the chance a fully agreeing panel is wrong; the last column is the confidence gap between correct and incorrect answers, which is the signal a clinician would use to discount an answer.

Architecture	unan.	$P(\text{wr.}\,|\,\text{un.})$	conf. gap
Single agent	—	—	**3.3**
Independent	66.2%	46.5%	2.8
Centralized	68.6%	**44.6%**	2.7
Decentralized	**98.3%**	49.9%	**1.8**
Hybrid	43.9%	59.7%	2.1

4.5 The arithmetic premise of consultation

Majority voting has value only when the votes are independent. Every result above is a consequence of how badly that premise fails.

A nine-member panel carries 1.3 independent opinions. We measure the pairwise error correlation φ between agent slots: for each pair of panel positions, the correlation of their per-item error indicators. Across all architectures at $n_a \geq 3$, $\varphi = 0.73$. The effective number of independent opinions,

$$N_{\text{eff}} = \frac{N}{1 + (N-1)\varphi}, \qquad (2)$$

is then 1.21 at $N{=}3$, 1.27 at $N{=}5$ and 1.31 at $N{=}9$. Tripling the panel adds 0.10 of an independent opinion. Specialty role prompts do not make one model into several physicians; they make it into one physician wearing several labels.

Discussion drives the effective panel size to one. Peer discussion raises φ from 0.751 before the exchange to $\mathbf{0.986}$ after it (paired $t = 20.9$, $p = 8 \times 10^{-7}$). At $\varphi = 0.986$ a six-agent panel has $N_{\text{eff}} = 1.01$. A multidisciplinary consultation, at the moment it reaches consensus, is informationally equivalent to one physician, and the vote taken at that moment is a vote among copies. This is the mechanism behind every safety result in §4.6: unanimity rises to 98% because the opinions have merged, not because they have converged on truth; the error rate given unanimity stays at a coin flip because merging does not add information; and confidence rises on wrong answers because N agents agreeing looks like corroboration when it is repetition.

Why panel size cannot predict anything. The nested model comparison found panel size alone at cross-validated $R^2 = -0.006$. The independence measurement explains why that is not a null result but a necessary one: over $N \in \{3, \ldots, 9\}$, N_{eff} moves from 1.21 to 1.31. The regressor that matters barely varies, so the regressor we can control cannot predict. Any design effort spent on how many agents to deploy is spent on the wrong variable.

What independence does not explain. Error correlation accounts for the size and consensus results; it does not account for the window. φ correlates only weakly with the single-doctor baseline ($r = -0.215$, $p = 0.018$: harder items do carry more correlated errors) and not at all with collaboration gain ($r = -0.083$, $p = 0.37$), and a model containing φ alone reaches $R^2 = 0.007$. The window and the independence collapse are two separate facts about medical consultation: the first says when a panel can help, the second says why it cannot help much, and why its agreement should not be read as evidence.

4.6 Clinical safety: what consultation does to confidence

Accuracy is not the only thing coordination changes. In medicine the more consequential question is what it does to the *signals a clinician would act on*. Three measurements, all pointing the same way.

Self-reported confidence barely discriminates right from wrong, and discussion makes it worse. Across all configurations, mean confidence on correct answers exceeds that on wrong answers by only 3.3 points for a single doctor (86.0 vs 82.6; AUC = 0.597). Peer discussion compresses that gap by 45%, to 1.8 points (88.4 vs 86.6; AUC = 0.577). The compression is not a rescaling: discussion raises confidence on *wrong* answers by 4.0 points while raising it on correct answers by 2.4. A panel that has deliberated is more sure of itself, and disproportionately so when it is mistaken. This is consistent with the threshold calibration we ran before the main experiments, where the best achievable referral gate on stated confidence reached a Youden J of only 0.071.

Discussion manufactures consensus without manufacturing correctness. Unanimity, defined as every panelist emitting the same valid option in the last round with two or more agents, rises from 66.2% under independent voting to 98.3% under peer discussion. Conditional accuracy does not follow: $P(\text{wrong} \mid \text{unanimous})$ is 46.5% for Independent and 49.9% for Decentralized. A unanimous medical panel is wrong about half the time in both cases; discussion simply produces far more unanimous panels. Centralized coordination is the only architecture that lowers the conditional error rate (44.6%), which is the error-containment role the orchestrator is supposed to play.

The most trustworthy-looking output is the least trustworthy. Taken together: consultation drives panels to near-total agreement, leaves the error rate given agreement at a coin flip, and raises confidence most on the answers that are wrong while shrinking the confidence gap that would let a reader tell the difference. The configuration a clinician would most readily act on, a unanimous and high-confidence multidisciplinary recommendation, carries essentially the same error rate as a divided one, and advertises itself more strongly. Accuracy gains of 6–8 pp inside the window are real, but they are purchased alongside a degradation in the calibration of the signal that would let a clinician catch the remaining errors.

5 Conclusion

Consultation works in human medicine because physicians fail differently. Our measurements say that language-model panels do not. Agents prompted into different specialties make the same mistakes ($\varphi = 0.73$), and discussion pushes them into near-perfect agreement ($\varphi = 0.986$), leaving a six-member consultation with 1.01 effective independent opinions. Every other result in this paper is a shadow of that one.

Panel size cannot matter, because it does not change what the panel knows. Across $N \in \{3, \ldots, 9\}$ the effective number of opinions moves from 1.21 to 1.31, and panel size alone predicts collaboration gain at cross-validated $R^2 = -0.006$. Consensus cannot be trusted, because it is agreement among copies. Unanimity reaches 98% while the error rate given unanimity stays at 50%, and the confidence signal that would let a clinician catch the difference is compressed by 45%. What remains is a narrow window of task difficulty, roughly 25–50% single-doctor accuracy. Inside it, coordination recovers enough complementary information to pay for itself. Outside it, coordination reliably does not.

Two design conclusions follow for medical deployments. First, the variable worth controlling is not how many agents but whether a competent one audits the others before the system commits: centralized coordination with a capable orchestrator produces the largest gain we measure, and the same architecture with a weak orchestrator produces the largest loss. Specialist capability matters $4.1\times$ more than the orchestrator's, so capability belongs on the agents doing the reasoning. Second, the deployment decision should be made by measuring the single-model baseline on the intended case mix before choosing an architecture at all. Outside the window, a budget-matched self-consistency control matches or beats every architecture we tested, at a fraction of the cost.

The broader caution is about what a consulting AI system advertises. A unanimous, high-confidence multidisciplinary recommendation is the output a clinician is most likely to act on, and in these experiments it is the output whose reliability is least legible from its own signals. Making panels agree is easy. Making them independent is the open problem.

6 Limitations and future works

Our tasks are multiple-choice examination items, not clinical encounters. They carry no history taking, no test ordering, no longitudinal follow-up, and the exact-match endpoint rewards the option-selection component of diagnosis only. The $T = 0$ property that makes the design clean also bounds what it licenses. Results here do not extrapolate to tool-using clinical agents.

All models are from a single vendor. The heterogeneity arm of the original design, which mixes vendors within a capability tier, is here restricted to mixing separately trained OpenAI families; genuine cross-vendor error decorrelation may be larger than what we measure.

Panels are nested by construction (§2.2), so the reported curves estimate accuracy as a routed panel is extended, not the expected accuracy of independently redrawn panels at each size. The coupling reduces variance on the architecture contrasts we care about and does not bias within-item paired tests, but a reader interested in the independent-panel estimand should read the curves accordingly.

The specialty router ranks by relevance, so larger panels append progressively less relevant specialists. "More experts" and "weaker experts" are therefore entangled in the specialty-role condition. We separate them with the generic-internist control, in which every panelist is an undifferentiated internist and panel size is the only thing that varies.

Difficulty strata are defined by the pass rate of mid-tier models, some of which also appear in the evaluation ladder. This inflates the level of the hard-stratum results but not the within-item slope across panel sizes, which is computed on identical items.

6.1 Success per 1K tokens

Token efficiency separates the architectures more sharply than accuracy does. Successes per 1,000 tokens: Hybrid 0.80, self-consistency 0.47, Independent 0.39, Decentralized 0.33, Centralized 0.19. The architecture that produced every significant accuracy gain in this study is also the least token-efficient, by $4.3\times$. Reported per question at list prices, a Centralized panel at $N{=}3$ costs \$1.4 per 1,000 MedXpertQA questions against \$0.21 for a single chain-of-thought call. That is a $6.7\times$ premium for $+7.6$ pp, and only inside the window. Outside it the same premium buys nothing.

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A Model capability index

Prior work scores models on a published Intelligence Index and reports that a *task-grounded* capability metric explains more variance ($R^2 = 0.413$) than the generic one ($R^2 = 0.373$). We therefore use the task-grounded form throughout: I is a model's mean single-doctor chain-of-thought accuracy across the three benchmarks. Table 4 gives the components. The three tiers span $I = 34.0$ to 59.2, and their single-benchmark baselines span 15.6% to 91.6%, crossing the 45% region from both sides. That crossing is what makes the capability-ceiling test possible at all.

B Domain complexity

B.1 Complexity metric construction

Domain complexity $D \in [0, 1]$ is the arithmetic mean of three complementary measures (Kim et al., 2026): the **performance ceiling** $1 - p_{\max}$, where $p_{\max}$ is the highest performance any evaluated system reaches; the **coefficient of variation** σ/μ of performance across all configurations, a scale-invariant measure of relative spread; and the **best-model baseline** $1 - p_{\text{best}}$, where p_{best} is the strongest single-model performance on the dataset.

B.2 Domain characterisation

Table 5 gives the components. MedQA is a low-complexity domain ($D = 0.097$); MedXpertQA and MedAgentsBench-hard are high-complexity and nearly indistinguishable ($D = 0.489$ and 0.478).

B.3 The critical threshold does not transfer

A critical complexity threshold at $D \approx 0.40$ has been reported for general-domain agents (Kim et al., 2026): below it multi-agent architectures yield net positive returns, above it coordination overhead consumes resources otherwise spent on reasoning. Our data do not support a benchmark-level threshold of any value. MedXpertQA ($D = 0.489$) and MedAgentsBench-hard ($D = 0.478$) differ by 0.011 in complexity yet by 3.5 pp in mean multi-agent gain (+3.09 pp versus −0.37 pp), giving opposite signs at effectively the same D. A benchmark-level statistic cannot separate them because the quantity that does separate them varies *within* each benchmark: on MedAgentsBench-hard the per-tier gains run −2.78, +1.02, +0.66 pp as P_{SA} moves from 20.4% through 32.4% to 44.4%, entering the window between the first tier and the second. Domain complexity is the right idea at the wrong granularity for medicine. The configuration-level baseline P_{SA} is what carries the signal.

C Datasets

MedXpertQA-Text (Zuo et al., 2025): 2,450 expert-level items, ten options each, tagged by 12 body systems and 3 task types. We draw a stratified sample interleaved across (body system × task) so that any prefix is approximately balanced, then take the first 250. **MedQA-USMLE** (Jin et al., 2021): 1,273 four-option

Table 4: Capability index and its components (single-doctor CoT accuracy, %).

Model	MedXpertQA	MedAgentsBench	MedQA	I (mean)	Reasoning
`gpt-4.1-nano`	15.6	20.4	66.0	**34.0**	—
`gpt-5-nano`	32.0	32.4	88.0	**50.8**	effort=low
`gpt-5-mini`	41.6	44.4	91.6	**59.2**	effort=low

Table 5: Domain complexity of the three medical benchmarks.

Benchmark	$1 - p_{\max}$	σ/μ	$1 - p_{\text{best}}$	D	mean MAS gain
MedXpertQA	0.508	0.376	0.584	**0.489**	+3.09 pp
MedAgentsBench-hard	0.532	0.346	0.556	**0.478**	-0.37 pp
MedQA (USMLE)	0.060	0.147	0.084	**0.097**	+0.07 pp

test items; randomly shuffled with a fixed seed, first 250. **MedAgentsBench-hard** (Tang et al., 2025): the contamination-screened hard split, 894 items after excluding the `medqa_5options` subset, stratified-interleaved across the ten source subsets, first 250. All samples are nested prefixes, so a larger sample re-uses every call of a smaller one.

Item difficulty is additionally tagged easy/medium/hard by the pass rate of three mid-tier models at k=5; on the MedXpertQA sample this yields 13/62/125 items.

D Implementation details

D.1 Technical infrastructure

All models are accessed through one OpenAI-compatible client with a content-addressed disk cache, exponential backoff with rate-limit-aware retry, a per-call JSONL ledger recording tokens, latency and cost, and a cross-process spend journal guarded by an OS file lock. Every LLM call is cached on the full request (model, system, messages, temperature, seed, max tokens, effort, JSON mode, tag), so a re-run is a replay.

D.2 Agent configuration

Single agents answer in one call. Independent panels deploy N specialists with synthesis-only aggregation. Centralized systems use one orchestrator over N sub-agents with a maximum of 2 orchestration rounds and one re-task per round. Decentralized systems run N agents through up to 3 discussion rounds, stopping early only on true unanimity. Hybrid systems combine a generalist triage gate with the specialist panel and escalate to full discussion when the panel has no majority. Agent temperature is 0.7 within panels, 0.3 for single-doctor baselines and for orchestrator/moderator turns; reasoning models are called at `effort=low` with a minimum completion budget of 1,600 tokens, since reasoning tokens are drawn from the same budget and a smaller cap returns empty output.

D.3 Prompt compilation

Every agent receives the same rendered item (stem, then options as `(A) ...`) and the same answer schema: JSON with `answer`, `confidence` (0–100) and `reason` (at most 45 words). Only the system prompt differs, naming the specialty. Peer context in communicating architectures is capped at a fixed 900-character budget, balanced-truncated across peers and *independent of* N, so a panel-size effect cannot be a context-length effect. Peer ordering is deterministically rotated per (item, round) so that the first-listed speaker is decorrelated from the top-ranked specialty.

D.4 Evaluation methodology

Sample sizes. 250 items per benchmark, 180 multi-agent configurations, 63,499 episodes. **Controls.** All architectures share the item rendering, answer schema, role-prompt template and decoding parameters; only coordination logic differs. Model weights are frozen. The primary endpoint is exact match against the dataset's own gold option key. No LLM judge appears anywhere in the evaluation path. Episodes that fail for infrastructure reasons carry an error status and are excluded from accuracy rather than scored as wrong, which would bias call-heavy architectures downward. Cost accounting is nominal: cached calls are charged at list price, so reported economics describe what a configuration would cost to run.

D.5 Information gain computation

Information gain is the Bayesian posterior variance reduction (Kim et al., 2026)

$$\Delta\mathcal{I} = \tfrac{1}{2}\log\frac{\mathrm{Var}[Y \mid \mathbf{s}_{\mathrm{pre}}]}{\mathrm{Var}[Y \mid \mathbf{s}_{\mathrm{post}}]}, \tag{3}$$

with $\mathrm{Var}[Y \mid \mathbf{s}] = \hat{p}(\mathbf{s})(1-\hat{p}(\mathbf{s}))$ and $Y \in \{0,1\}$ the success indicator. They estimate $\hat{p}$ from $K{=}10$ sampled reasoning traces at temperature 0.7. In multiple-choice consultation the panel already supplies such a sample: $\hat{p}$ is the fraction of the round's opinions that name the gold option, Laplace smoothed as $(k{+}1)/(n{+}2)$. Smoothing rather than clipping matters. A Centralized panel's final round contains only the orchestrator, and clipping would pin $\hat{p}$ at 0.5 and force $\Delta\mathcal{I}$ negative by construction.

E Complete results grid

F Additional results

F.1 Where capability belongs inside a panel

Specialist capability matters four times more than the orchestrator's. Replacing the whole panel with weak models costs 34.4 pp (49.2% to 14.8%). Splitting that substitution isolates where the loss comes from. Downgrading only the *orchestrator*, keeping strong specialists, costs 3.6 pp, so the panel retains 90% of the homogeneous-high span. Downgrading only the *specialists*, keeping a strong orchestrator, costs 14.8 pp, retaining 57%. A strong attending physician recovers more than half of what a weak panel loses, but a weak attending physician barely dents a strong one. The asymmetry is $4.1\times$. In a centralized clinical panel, capability should therefore be spent on the agents doing the reasoning rather than on the agent doing the auditing.

An orchestrator too weak to audit is worse than no orchestrator at all. The largest single degradation anywhere in this study is a Centralized panel with a weak model on MedQA at $P_{\mathrm{SA}} = 66.0\%$: -13.60 pp, $p = 10^{-5}$. Inspection of those episodes shows the mechanism directly. The orchestrator overrides a correct specialist majority. This is the error-amplification term in its most extreme form. It is also the counterpart to the previous result. The audit is what makes centralized coordination work, so an auditor that cannot audit converts the architecture's strength into its worst liability.

Mixed-capability panels land between their components, with no emergent benefit. A panel of one strong, one mid and one weak specialist reaches 40.8% under Centralized and 44.0% under Decentralized, against 49.2% and 47.6% for homogeneous-high. Both sit below the strong homogeneous panel and well above the weak one, roughly where a capability-weighted average would put them. We find no medical analogue of the heterogeneous-mixing advantage reported for one model family (Kim et al., 2026), in which a low-capability orchestrator with high-capability sub-agents outperformed a homogeneous high-capability panel by 31%. In our data that configuration is the second-best Centralized cell but still 7% below homogeneous-high, directionally matching what they observe for OpenAI models, where heterogeneous centralized configurations degrade.

F.2 Where in medicine collaboration pays

Collaboration helps diagnosis most, and it is the only task type significant at both tiers. MedXpertQA labels each item as Diagnosis, Treatment, or Basic Science. Splitting the window-resident cells by label, the best panel gains +9.5 pp on Diagnosis at `gpt-5-nano` ($p = 0.008$) and +10.7 pp at `gpt-5-mini` ($p = 0.004$), against +7.1 and +8.3 pp on Treatment and +7.3 and +8.5 pp on Basic Science. Diagnosis is the only category significant at both tiers. This is the pattern clinical intuition predicts: differential diagnosis is the medical task that most resembles pooling independent perspectives, and it is where pooling pays.

The difficulty stratification is where the window shows up within a single benchmark. On MedXpertQA at `gpt-5-mini`, the best configuration gains +7.7 pp on easy items ($n{=}13$), +6.5 pp on medium ($n{=}62$) and +9.6 pp on hard ($n{=}125$). Gains are positive in every stratum for this tier, which sits inside the window; the sign flips only for tiers whose baseline falls outside it. Item difficulty and configuration baseline are two views of the same quantity. The window is defined on the view that can be measured before deployment.

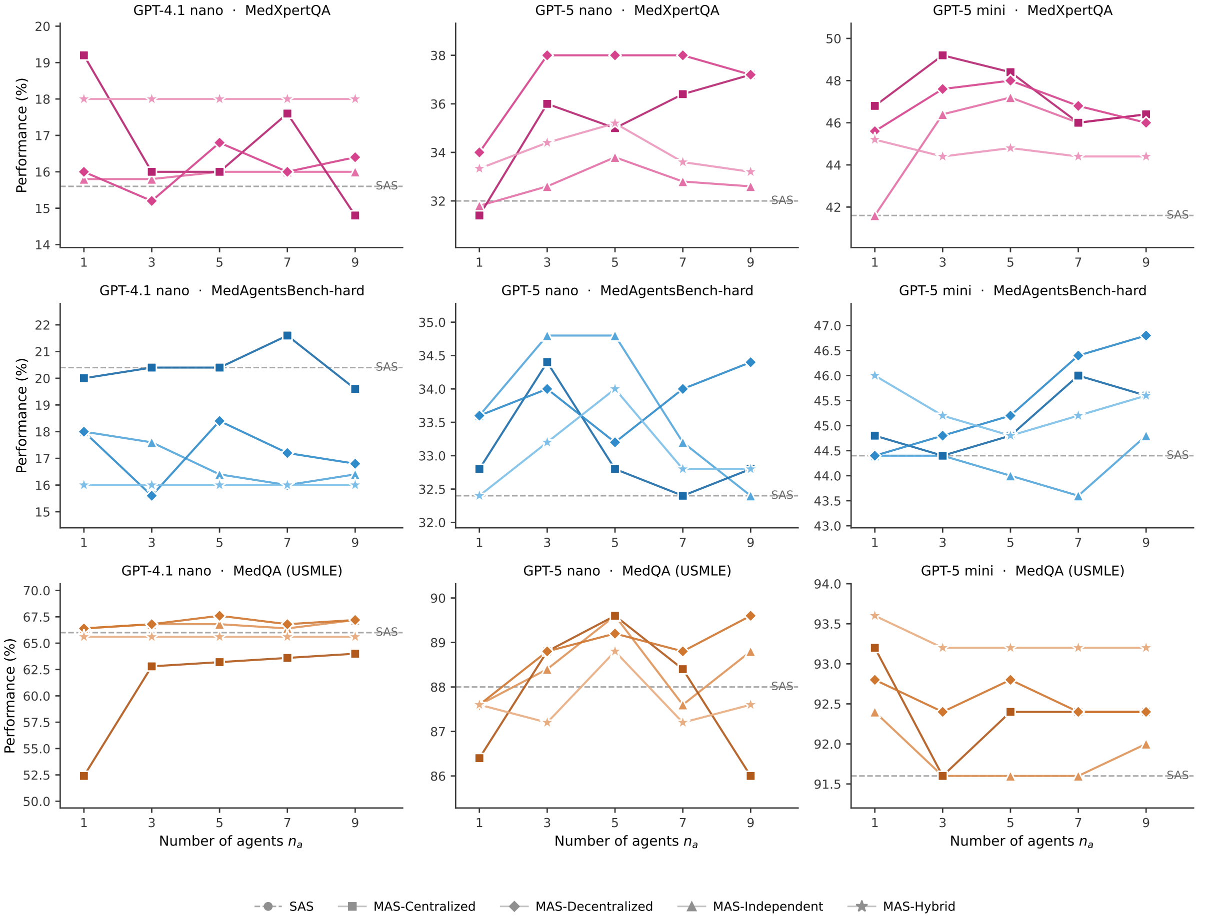

Figure 3: **Number-of-agents scaling.** Performance against panel size $n_a \in \{1, 3, 5, 7, 9\}$ for every architecture, model and benchmark. Dashed grey: single-agent baseline. Dotted: budget-matched self-consistency. Where gains occur they are captured by $n_a = 3$.

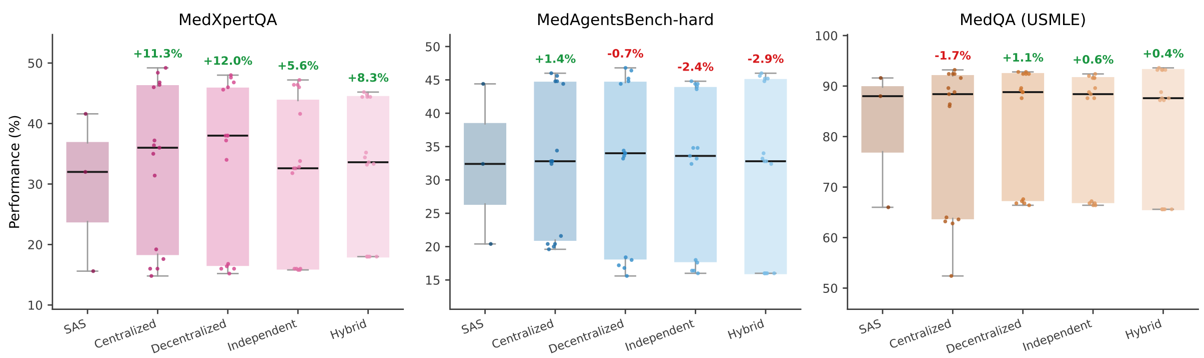

Figure 4: **Single-agent versus multi-agent performance distributions.** Box plots over all configurations of each architecture, with individual configurations overlaid. Annotations give the change in mean relative to the single-agent baseline.

Table 6: Complete results grid (1 of 4). All 180 multi-agent configurations and their single-model controls.

Tier	Benchmark	Architecture	n_a	Acc. (%)	95% CI	USD/1k
T1	MedAgentsBench	Centralized	1	20.0	[15.5, 25.4]	0.14
T1	MedAgentsBench	Centralized	3	20.4	[15.9, 25.8]	0.25
T1	MedAgentsBench	Centralized	5	20.4	[15.9, 25.8]	0.36
T1	MedAgentsBench	Centralized	7	21.6	[16.9, 27.1]	0.47
T1	MedAgentsBench	Centralized	9	19.6	[15.2, 25.0]	0.56
T1	MedAgentsBench	SAS CoT	1	20.4	[15.9, 25.8]	0.08
T1	MedAgentsBench	Decentralized	1	18.0	[13.7, 23.2]	0.05
T1	MedAgentsBench	Decentralized	3	15.6	[11.6, 20.6]	0.21
T1	MedAgentsBench	Decentralized	5	18.4	[14.1, 23.7]	0.43
T1	MedAgentsBench	Decentralized	7	17.2	[13.0, 22.4]	0.61
T1	MedAgentsBench	Decentralized	9	16.8	[12.7, 21.9]	0.88
T1	MedAgentsBench	Independent	1	18.0	[13.7, 23.2]	0.05
T1	MedAgentsBench	Independent	3	17.6	[13.4, 22.8]	0.14
T1	MedAgentsBench	Independent	5	16.4	[12.3, 21.5]	0.24
T1	MedAgentsBench	Independent	7	16.0	[12.0, 21.1]	0.33
T1	MedAgentsBench	Independent	9	16.4	[12.3, 21.5]	0.43
T1	MedAgentsBench	Self-consist.	1	20.0	[15.5, 25.4]	0.05
T1	MedAgentsBench	Self-consist.	3	18.8	[14.4, 24.1]	0.09
T1	MedAgentsBench	Self-consist.	5	19.2	[14.8, 24.5]	0.13
T1	MedAgentsBench	Self-consist.	9	18.0	[13.7, 23.2]	0.25
T1	MedAgentsBench	Self-consist.	15	18.4	[14.1, 23.7]	0.37
T1	MedAgentsBench	Hybrid	1	16.0	[12.0, 21.1]	0.05
T1	MedAgentsBench	Hybrid	3	16.0	[12.0, 21.1]	0.05
T1	MedAgentsBench	Hybrid	5	16.0	[12.0, 21.1]	0.05
T1	MedAgentsBench	Hybrid	7	16.0	[12.0, 21.1]	0.05
T1	MedAgentsBench	Hybrid	9	16.0	[12.0, 21.1]	0.05
T1	MedAgentsBench	SAS zero-shot	1	16.8	[12.7, 21.9]	0.03
T1	MedQA	Centralized	1	52.4	[46.2, 58.5]	0.14
T1	MedQA	Centralized	3	62.8	[56.7, 68.6]	0.25
T1	MedQA	Centralized	5	63.2	[57.1, 68.9]	0.36
T1	MedQA	Centralized	7	63.6	[57.5, 69.3]	0.46
T1	MedQA	Centralized	9	64.0	[57.9, 69.7]	0.57
T1	MedQA	SAS CoT	1	66.0	[59.9, 71.6]	0.08
T1	MedQA	Decentralized	1	66.4	[60.3, 72.0]	0.05
T1	MedQA	Decentralized	3	66.8	[60.7, 72.3]	0.20
T1	MedQA	Decentralized	5	67.6	[61.6, 73.1]	0.35
T1	MedQA	Decentralized	7	66.8	[60.7, 72.3]	0.56
T1	MedQA	Decentralized	9	67.2	[61.2, 72.7]	0.75
T1	MedQA	Independent	1	66.4	[60.3, 72.0]	0.05
T1	MedQA	Independent	3	66.8	[60.7, 72.3]	0.15
T1	MedQA	Independent	5	66.8	[60.7, 72.3]	0.25
T1	MedQA	Independent	7	66.4	[60.3, 72.0]	0.35
T1	MedQA	Independent	9	67.2	[61.2, 72.7]	0.45
T1	MedQA	Self-consist.	1	68.4	[62.4, 73.8]	0.05
T1	MedQA	Self-consist.	3	65.6	[59.5, 71.2]	0.10
T1	MedQA	Self-consist.	5	65.6	[59.5, 71.2]	0.14
T1	MedQA	Self-consist.	9	66.0	[59.9, 71.6]	0.26
T1	MedQA	Self-consist.	15	65.6	[59.5, 71.2]	0.40
T1	MedQA	Hybrid	1	65.6	[59.5, 71.2]	0.05
T1	MedQA	Hybrid	3	65.6	[59.5, 71.2]	0.05
T1	MedQA	Hybrid	5	65.6	[59.5, 71.2]	0.05
T1	MedQA	Hybrid	7	65.6	[59.5, 71.2]	0.05
T1	MedQA	Hybrid	9	65.6	[59.5, 71.2]	0.05
T1	MedQA	SAS zero-shot	1	66.4	[60.3, 72.0]	0.03
T1	MedXpertQA	Centralized	1	19.2	[14.8, 24.5]	0.17
T1	MedXpertQA	Centralized	3	16.0	[12.0, 21.1]	0.28
T1	MedXpertQA	Centralized	5	16.0	[12.0, 21.1]	0.40
T1	MedXpertQA	Centralized	7	17.6	[13.4, 22.8]	0.52
T1	MedXpertQA	Centralized	9	14.8	[10.9, 19.7]	0.62
T1	MedXpertQA	SAS CoT	1	15.6	[11.6, 20.6]	0.09
T1	MedXpertQA	Decentralized	1	16.0	[12.0, 21.1]	0.05
T1	MedXpertQA	Decentralized	3	15.2	[11.3, 20.2]	0.25

Table 7: Complete results grid (2 of 4). All 180 multi-agent configurations and their single-model controls.

Tier	Benchmark	Architecture	n_a	Acc. (%)	95% CI	USD/1k
T1	MedXpertQA	Decentralized	5	16.8	[12.7, 21.9]	0.50
T1	MedXpertQA	Decentralized	7	16.0	[12.0, 21.1]	0.74
T1	MedXpertQA	Decentralized	9	16.4	[12.3, 21.5]	1.05
T1	MedXpertQA	Independent	1	15.8	[12.9, 19.3]	0.05
T1	MedXpertQA	Independent	3	15.8	[12.9, 19.3]	0.16
T1	MedXpertQA	Independent	5	16.0	[12.0, 21.1]	0.27
T1	MedXpertQA	Independent	7	16.0	[12.0, 21.1]	0.38
T1	MedXpertQA	Independent	9	16.0	[12.0, 21.1]	0.48
T1	MedXpertQA	Self-consist.	1	16.0	[12.0, 21.1]	0.06
T1	MedXpertQA	Self-consist.	3	16.4	[12.3, 21.5]	0.10
T1	MedXpertQA	Self-consist.	5	16.0	[12.0, 21.1]	0.14
T1	MedXpertQA	Self-consist.	9	15.6	[11.6, 20.6]	0.27
T1	MedXpertQA	Self-consist.	15	16.0	[12.0, 21.1]	0.40
T1	MedXpertQA	Hybrid	1	18.0	[13.7, 23.2]	0.05
T1	MedXpertQA	Hybrid	3	18.0	[13.7, 23.2]	0.05
T1	MedXpertQA	Hybrid	5	18.0	[13.7, 23.2]	0.05
T1	MedXpertQA	Hybrid	7	18.0	[13.7, 23.2]	0.06
T1	MedXpertQA	Hybrid	9	18.0	[13.7, 23.2]	0.06
T1	MedXpertQA	SAS zero-shot	1	16.4	[12.3, 21.5]	0.03
T2	MedAgentsBench	Centralized	1	32.8	[27.3, 38.8]	0.62
T2	MedAgentsBench	Centralized	3	34.4	[28.8, 40.5]	0.93
T2	MedAgentsBench	Centralized	5	32.8	[27.3, 38.8]	1.25
T2	MedAgentsBench	Centralized	7	32.4	[26.9, 38.4]	1.59
T2	MedAgentsBench	Centralized	9	32.8	[27.3, 38.8]	1.91
T2	MedAgentsBench	SAS CoT	1	32.4	[26.9, 38.4]	0.19
T2	MedAgentsBench	Decentralized	1	33.6	[28.0, 39.7]	0.16
T2	MedAgentsBench	Decentralized	3	34.0	[28.4, 40.1]	0.73
T2	MedAgentsBench	Decentralized	5	33.2	[27.7, 39.3]	1.34
T2	MedAgentsBench	Decentralized	7	34.0	[28.4, 40.1]	2.11
T2	MedAgentsBench	Decentralized	9	34.4	[28.8, 40.5]	2.92
T2	MedAgentsBench	Independent	1	33.6	[28.0, 39.7]	0.16
T2	MedAgentsBench	Independent	3	34.8	[29.2, 40.9]	0.48
T2	MedAgentsBench	Independent	5	34.8	[29.2, 40.9]	0.79
T2	MedAgentsBench	Independent	7	33.2	[27.7, 39.3]	1.12
T2	MedAgentsBench	Independent	9	32.4	[26.9, 38.4]	1.44
T2	MedAgentsBench	Self-consist.	1	33.2	[27.7, 39.3]	0.18
T2	MedAgentsBench	Self-consist.	3	32.0	[26.5, 38.0]	0.50
T2	MedAgentsBench	Self-consist.	5	33.6	[28.0, 39.7]	0.83
T2	MedAgentsBench	Self-consist.	9	32.8	[27.3, 38.8]	1.49
T2	MedAgentsBench	Self-consist.	15	31.2	[25.8, 37.2]	2.47
T2	MedAgentsBench	Hybrid	1	32.4	[26.9, 38.4]	0.26
T2	MedAgentsBench	Hybrid	3	33.2	[27.7, 39.3]	0.49
T2	MedAgentsBench	Hybrid	5	34.0	[28.4, 40.1]	0.73
T2	MedAgentsBench	Hybrid	7	32.8	[27.3, 38.8]	0.94
T2	MedAgentsBench	Hybrid	9	32.8	[27.3, 38.8]	1.16
T2	MedAgentsBench	SAS zero-shot	1	33.2	[27.7, 39.3]	0.13
T2	MedQA	Centralized	1	86.4	[81.6, 90.1]	0.47
T2	MedQA	Centralized	3	88.8	[84.3, 92.1]	0.67
T2	MedQA	Centralized	5	89.6	[85.2, 92.8]	0.95
T2	MedQA	Centralized	7	88.4	[83.8, 91.8]	1.23
T2	MedQA	Centralized	9	86.0	[81.2, 89.8]	1.51
T2	MedQA	SAS CoT	1	88.0	[83.4, 91.5]	0.16
T2	MedQA	Decentralized	1	87.6	[82.9, 91.1]	0.13
T2	MedQA	Decentralized	3	88.8	[84.3, 92.1]	0.50
T2	MedQA	Decentralized	5	89.2	[84.7, 92.5]	0.86
T2	MedQA	Decentralized	7	88.8	[84.3, 92.1]	1.26
T2	MedQA	Decentralized	9	89.6	[85.2, 92.8]	1.75
T2	MedQA	Independent	1	87.6	[82.9, 91.1]	0.13
T2	MedQA	Independent	3	88.4	[83.8, 91.8]	0.39
T2	MedQA	Independent	5	89.6	[85.2, 92.8]	0.65
T2	MedQA	Independent	7	87.6	[82.9, 91.1]	0.90
T2	MedQA	Independent	9	88.8	[84.3, 92.1]	1.16

Table 8: Complete results grid (3 of 4). All 180 multi-agent configurations and their single-model controls.

Tier	Benchmark	Architecture	n_a	Acc. (%)	95% CI	USD/1k
T2	MedQA	Self-consist.	1	84.4	[79.4, 88.4]	0.15
T2	MedQA	Self-consist.	3	86.8	[82.0, 90.4]	0.41
T2	MedQA	Self-consist.	5	88.0	[83.4, 91.5]	0.67
T2	MedQA	Self-consist.	9	87.6	[82.9, 91.1]	1.21
T2	MedQA	Self-consist.	15	88.4	[83.8, 91.8]	2.01
T2	MedQA	Hybrid	1	87.6	[82.9, 91.1]	0.17
T2	MedQA	Hybrid	3	87.2	[82.5, 90.8]	0.26
T2	MedQA	Hybrid	5	88.8	[84.3, 92.1]	0.35
T2	MedQA	Hybrid	7	87.2	[82.5, 90.8]	0.43
T2	MedQA	Hybrid	9	87.6	[82.9, 91.1]	0.53
T2	MedQA	SAS zero-shot	1	87.2	[82.5, 90.8]	0.09
T2	MedXpertQA	Centralized	1	31.4	[27.5, 35.6]	0.73
T2	MedXpertQA	Centralized	3	36.0	[31.9, 40.3]	1.02
T2	MedXpertQA	Centralized	5	35.0	[30.9, 39.3]	1.38
T2	MedXpertQA	Centralized	7	36.4	[30.7, 42.5]	1.80
T2	MedXpertQA	Centralized	9	37.2	[31.4, 43.3]	2.17
T2	MedXpertQA	SAS CoT	1	32.0	[26.5, 38.0]	0.20
T2	MedXpertQA	Decentralized	1	34.0	[28.4, 40.1]	0.19
T2	MedXpertQA	Decentralized	3	38.0	[32.2, 44.2]	0.91
T2	MedXpertQA	Decentralized	5	38.0	[32.2, 44.2]	1.66
T2	MedXpertQA	Decentralized	7	38.0	[32.2, 44.2]	2.46
T2	MedXpertQA	Decentralized	9	37.2	[31.4, 43.3]	3.28
T2	MedXpertQA	Independent	1	31.8	[27.9, 36.0]	0.19
T2	MedXpertQA	Independent	3	32.6	[28.6, 36.8]	0.56
T2	MedXpertQA	Independent	5	33.8	[29.8, 38.1]	0.93
T2	MedXpertQA	Independent	7	32.8	[28.8, 37.0]	1.30
T2	MedXpertQA	Independent	9	32.6	[28.6, 36.8]	1.67
T2	MedXpertQA	Self-consist.	1	32.8	[27.3, 38.8]	0.20
T2	MedXpertQA	Self-consist.	3	36.0	[30.3, 42.1]	0.57
T2	MedXpertQA	Self-consist.	5	34.8	[29.2, 40.9]	0.94
T2	MedXpertQA	Self-consist.	9	35.2	[29.5, 41.3]	1.70
T2	MedXpertQA	Self-consist.	15	34.4	[28.8, 40.5]	2.81
T2	MedXpertQA	Hybrid	1	33.3	[27.8, 39.4]	0.29
T2	MedXpertQA	Hybrid	3	34.4	[28.8, 40.5]	0.60
T2	MedXpertQA	Hybrid	5	35.2	[29.5, 41.3]	0.89
T2	MedXpertQA	Hybrid	7	33.6	[28.0, 39.7]	1.18
T2	MedXpertQA	Hybrid	9	33.2	[27.7, 39.3]	1.56
T2	MedXpertQA	SAS zero-shot	1	32.4	[26.9, 38.4]	0.15
T3	MedAgentsBench	Centralized	1	44.8	[38.8, 51.0]	1.54
T3	MedAgentsBench	Centralized	3	44.4	[38.4, 50.6]	2.40
T3	MedAgentsBench	Centralized	5	44.8	[38.8, 51.0]	3.51
T3	MedAgentsBench	Centralized	7	46.0	[39.9, 52.2]	4.60
T3	MedAgentsBench	Centralized	9	45.6	[39.5, 51.8]	5.76
T3	MedAgentsBench	SAS CoT	1	44.4	[38.4, 50.6]	0.68
T3	MedAgentsBench	Decentralized	1	44.4	[38.4, 50.6]	0.56
T3	MedAgentsBench	Decentralized	3	44.8	[38.8, 51.0]	2.11
T3	MedAgentsBench	Decentralized	5	45.2	[39.1, 51.4]	3.88
T3	MedAgentsBench	Decentralized	7	46.4	[40.3, 52.6]	5.99
T3	MedAgentsBench	Decentralized	9	46.8	[40.7, 53.0]	7.80
T3	MedAgentsBench	Independent	1	44.4	[38.4, 50.6]	0.55
T3	MedAgentsBench	Independent	3	44.4	[38.4, 50.6]	1.61
T3	MedAgentsBench	Independent	5	44.0	[38.0, 50.2]	2.67
T3	MedAgentsBench	Independent	7	43.6	[37.6, 49.8]	3.72
T3	MedAgentsBench	Independent	9	44.8	[38.8, 51.0]	4.77
T3	MedAgentsBench	Self-consist.	1	43.6	[37.6, 49.8]	0.62
T3	MedAgentsBench	Self-consist.	3	44.4	[38.4, 50.6]	1.73
T3	MedAgentsBench	Self-consist.	5	44.4	[38.4, 50.6]	2.84
T3	MedAgentsBench	Self-consist.	9	44.0	[38.0, 50.2]	5.12
T3	MedAgentsBench	Self-consist.	15	44.4	[38.4, 50.6]	8.43
T3	MedAgentsBench	Hybrid	1	46.0	[39.9, 52.2]	0.62
T3	MedAgentsBench	Hybrid	3	45.2	[39.1, 51.4]	0.78
T3	MedAgentsBench	Hybrid	5	44.8	[38.8, 51.0]	0.99

Table 9: Complete results grid (4 of 4). All 180 multi-agent configurations and their single-model controls.

Tier	Benchmark	Architecture	n_a	Acc. (%)	95% CI	USD/1k
T3	MedAgentsBench	Hybrid	7	45.2	[39.1, 51.4]	1.09
T3	MedAgentsBench	Hybrid	9	45.6	[39.5, 51.8]	1.25
T3	MedAgentsBench	SAS zero-shot	1	46.0	[39.9, 52.2]	0.39
T3	MedQA	Centralized	1	93.2	[89.4, 95.7]	1.07
T3	MedQA	Centralized	3	91.6	[87.5, 94.4]	2.00
T3	MedQA	Centralized	5	92.4	[88.4, 95.1]	2.84
T3	MedQA	Centralized	7	92.4	[88.4, 95.1]	3.80
T3	MedQA	Centralized	9	92.4	[88.4, 95.1]	4.66
T3	MedQA	SAS CoT	1	91.6	[87.5, 94.4]	0.57
T3	MedQA	Decentralized	1	92.8	[88.9, 95.4]	0.44
T3	MedQA	Decentralized	3	92.4	[88.4, 95.1]	1.50
T3	MedQA	Decentralized	5	92.8	[88.9, 95.4]	2.62
T3	MedQA	Decentralized	7	92.4	[88.4, 95.1]	3.70
T3	MedQA	Decentralized	9	92.4	[88.4, 95.1]	4.70
T3	MedQA	Independent	1	92.4	[88.4, 95.1]	0.43
T3	MedQA	Independent	3	91.6	[87.5, 94.4]	1.29
T3	MedQA	Independent	5	91.6	[87.5, 94.4]	2.15
T3	MedQA	Independent	7	91.6	[87.5, 94.4]	3.01
T3	MedQA	Independent	9	92.0	[88.0, 94.8]	3.86
T3	MedQA	Self-consist.	1	92.0	[88.0, 94.8]	0.52
T3	MedQA	Self-consist.	3	92.8	[88.9, 95.4]	1.41
T3	MedQA	Self-consist.	5	93.2	[89.4, 95.7]	2.30
T3	MedQA	Self-consist.	9	94.0	[90.3, 96.3]	4.15
T3	MedQA	Self-consist.	15	93.6	[89.9, 96.0]	6.78
T3	MedQA	Hybrid	1	93.6	[89.9, 96.0]	0.43
T3	MedQA	Hybrid	3	93.2	[89.4, 95.7]	0.48
T3	MedQA	Hybrid	5	93.2	[89.4, 95.7]	0.52
T3	MedQA	Hybrid	7	93.2	[89.4, 95.7]	0.56
T3	MedQA	Hybrid	9	93.2	[89.4, 95.7]	0.60
T3	MedQA	SAS zero-shot	1	94.4	[90.8, 96.6]	0.27
T3	MedXpertQA	Centralized	1	46.8	[40.7, 53.0]	1.53
T3	MedXpertQA	Centralized	3	49.2	[43.1, 55.4]	2.48
T3	MedXpertQA	Centralized	5	48.4	[42.3, 54.6]	3.68
T3	MedXpertQA	Centralized	7	46.0	[39.9, 52.2]	4.89
T3	MedXpertQA	Centralized	9	46.4	[40.3, 52.6]	6.07
T3	MedXpertQA	SAS CoT	1	41.6	[35.7, 47.8]	0.72
T3	MedXpertQA	Decentralized	1	45.6	[39.5, 51.8]	0.58
T3	MedXpertQA	Decentralized	3	47.6	[41.5, 53.8]	2.33
T3	MedXpertQA	Decentralized	5	48.0	[41.9, 54.2]	4.39
T3	MedXpertQA	Decentralized	7	46.8	[40.7, 53.0]	6.65
T3	MedXpertQA	Decentralized	9	46.0	[39.9, 52.2]	9.14
T3	MedXpertQA	Independent	1	41.6	[37.4, 46.0]	0.59
T3	MedXpertQA	Independent	3	46.4	[40.3, 52.6]	1.71
T3	MedXpertQA	Independent	5	47.2	[41.1, 53.4]	2.84
T3	MedXpertQA	Independent	7	46.0	[39.9, 52.2]	3.97
T3	MedXpertQA	Independent	9	46.4	[40.3, 52.6]	5.11
T3	MedXpertQA	Self-consist.	1	42.4	[36.4, 48.6]	0.67
T3	MedXpertQA	Self-consist.	3	43.2	[37.2, 49.4]	1.83
T3	MedXpertQA	Self-consist.	5	43.2	[37.2, 49.4]	3.00
T3	MedXpertQA	Self-consist.	9	46.0	[39.9, 52.2]	5.41
T3	MedXpertQA	Self-consist.	15	47.6	[41.5, 53.8]	8.87
T3	MedXpertQA	Hybrid	1	45.2	[39.1, 51.4]	0.61
T3	MedXpertQA	Hybrid	3	44.4	[38.4, 50.6]	0.71
T3	MedXpertQA	Hybrid	5	44.8	[38.8, 51.0]	0.87
T3	MedXpertQA	Hybrid	7	44.4	[38.4, 50.6]	0.93
T3	MedXpertQA	Hybrid	9	44.4	[38.4, 50.6]	1.20
T3	MedXpertQA	SAS zero-shot	1	43.2	[37.2, 49.4]	0.46

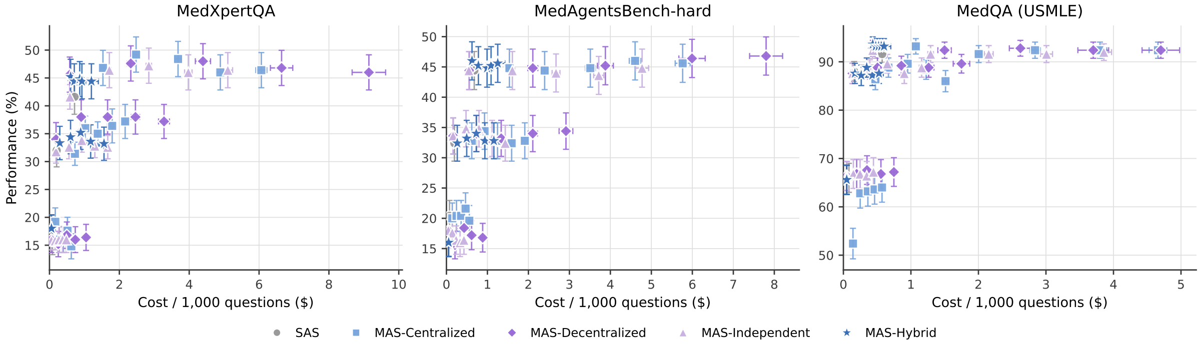

Figure 5: **Cost–performance trade-offs.** Mean performance against nominal cost per 1,000 questions, with SEM error bars on both axes. Each point is one configuration.

Table 10: Main results. For each capability tier and benchmark: the single-agent baseline, then each architecture at its best panel size, and the gain of the best architecture over the single agent. The complete 180-configuration grid is Table 6.

Tier	Benchmark	SAS	Best multi-agent variant (accuracy % / n_a)				Gain
		CoT	Independ.	Central.	Decentr.	Hybrid	best
T1	MedXpertQA	15.6	16.0 / 9	19.2 / 1	16.8 / 5	18.0 / 9	+3.6
T2	MedXpertQA	32.0	33.8 / 5	37.2 / 9	38.0 / 7	35.2 / 5	+6.0
T3	MedXpertQA	41.6	47.2 / 5	49.2 / 3	48.0 / 5	45.2 / 1	+7.6
T1	MedAgentsBench	20.4	18.0 / 1	21.6 / 7	18.4 / 5	16.0 / 9	+1.2
T2	MedAgentsBench	32.4	34.8 / 5	34.4 / 3	34.4 / 9	34.0 / 5	+2.4
T3	MedAgentsBench	44.4	44.8 / 9	46.0 / 7	46.8 / 9	46.0 / 1	+2.4
T1	MedQA	66.0	67.2 / 9	64.0 / 9	67.6 / 5	65.6 / 9	+1.6
T2	MedQA	88.0	89.6 / 5	89.6 / 5	89.6 / 9	88.8 / 5	+1.6
T3	MedQA	91.6	92.4 / 1	93.2 / 1	92.8 / 5	93.6 / 1	+2.0

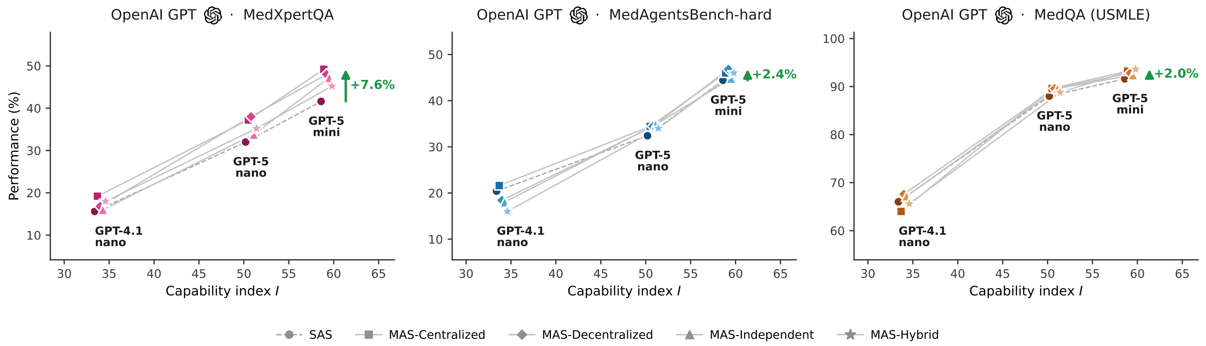

Figure 6: **Agent scaling across model capability and system architecture.** Performance against the task-grounded capability index I (mean single-doctor accuracy across the three benchmarks), for each architecture, on each benchmark. Each architecture is shown at its best-performing panel size. Dashed grey: single-agent baseline. Green arrows give the gain of the best architecture over the single agent at the strongest model.

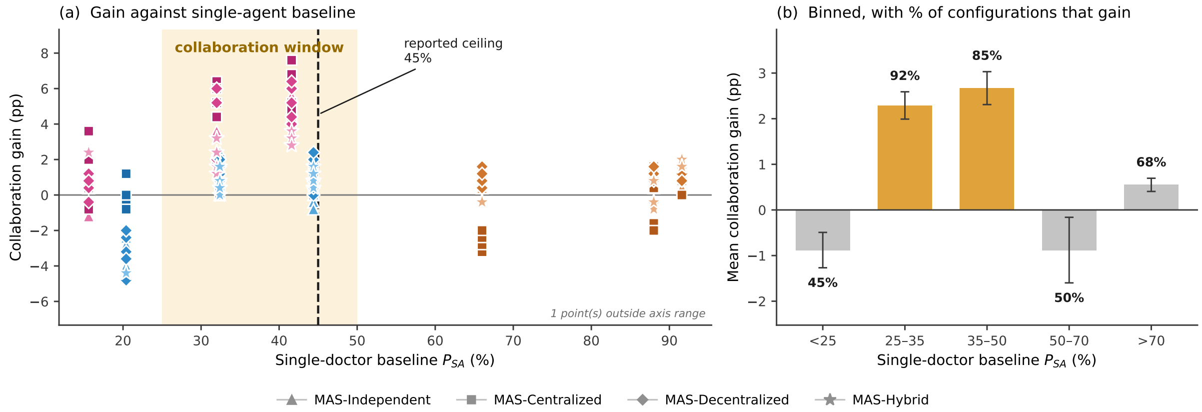

Figure 7: **Collaboration pays only inside a window of task difficulty.** (a) Collaboration gain against the single-doctor baseline P_{SA} for every multi-agent configuration; The shaded band marks the 25–50% window; the dashed line is the previously reported 45% capability ceiling. (b) The same data binned; annotations give the fraction of configurations in each bin that improve on the single doctor.

Table 11: Capability ladder, measured on 40 items per benchmark with single-doctor CoT.

Tier	Model	Effort	MedXpertQA	MedQA	USD / 1k items
T1	`gpt-4.1-nano`	—	25.0%	75.0%	0.09
T2	`gpt-5-nano`	low	37.5%	95.0%	0.17
T3	`gpt-5-mini`	low	55.0%	97.5%	0.64

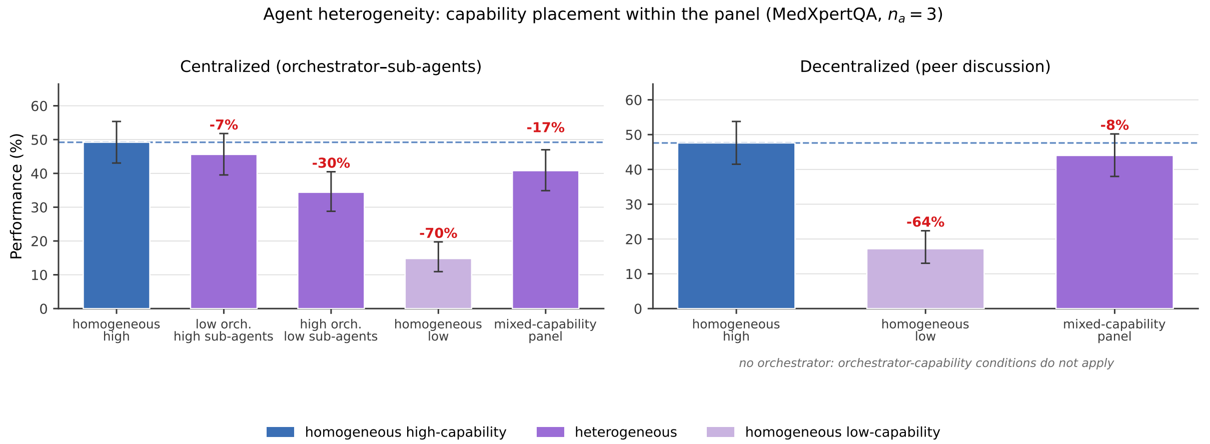

Figure 8: **Agent heterogeneity.** Performance of Centralized and Decentralized panels at n_a=3 on MedXpertQA under five capability placements. High capability is `gpt-5-mini`, low is `gpt-4.1-nano`. Annotations give the change relative to the homogeneous high-capability panel (dashed line). Decentralized coordination has no orchestrator, so the two orchestrator-capability conditions are structurally undefined there and are omitted.

Table 12: Agent heterogeneity at n_a=3 on MedXpertQA. High capability is `gpt-5-mini`, low is `gpt-4.1-nano`. "Span retained" is the fraction of the homogeneous-high minus homogeneous-low range that the configuration preserves. Decentralized coordination has no orchestrator, so those conditions are structurally undefined.

Capability placement	Centralized		Decentralized	
	accuracy	span retained	accuracy	span retained
Homogeneous high	49.2%	100%	47.6%	100%
Low orchestrator, high agents	45.6%	**90%**	—	—
High orchestrator, low agents	34.4%	**57%**	—	—
Mixed-capability panel	40.8%	76%	44.0%	88%
Homogeneous low	14.8%	0%	17.2%	0%

Table 13: Collaboration gain by clinical task type on MedXpertQA, for the two tiers inside the window. Gain is the best panel configuration over the single doctor on the same items; p from paired McNemar.

Tier	Task type	n	single doctor	best panel	gain
`gpt-5-nano`	Diagnosis	84	33.3%	42.9%	+9.52 pp**
	Treatment	84	35.7%	42.9%	+7.14 pp
	Basic Science	82	26.8%	34.1%	+7.32 pp
`gpt-5-mini`	Diagnosis	84	38.1%	48.8%	+10.71 pp**
	Treatment	84	44.0%	52.4%	+8.33 pp*
	Basic Science	82	42.7%	51.2%	+8.54 pp*

$^{*}p < 0.05$, $^{**}p < 0.01$